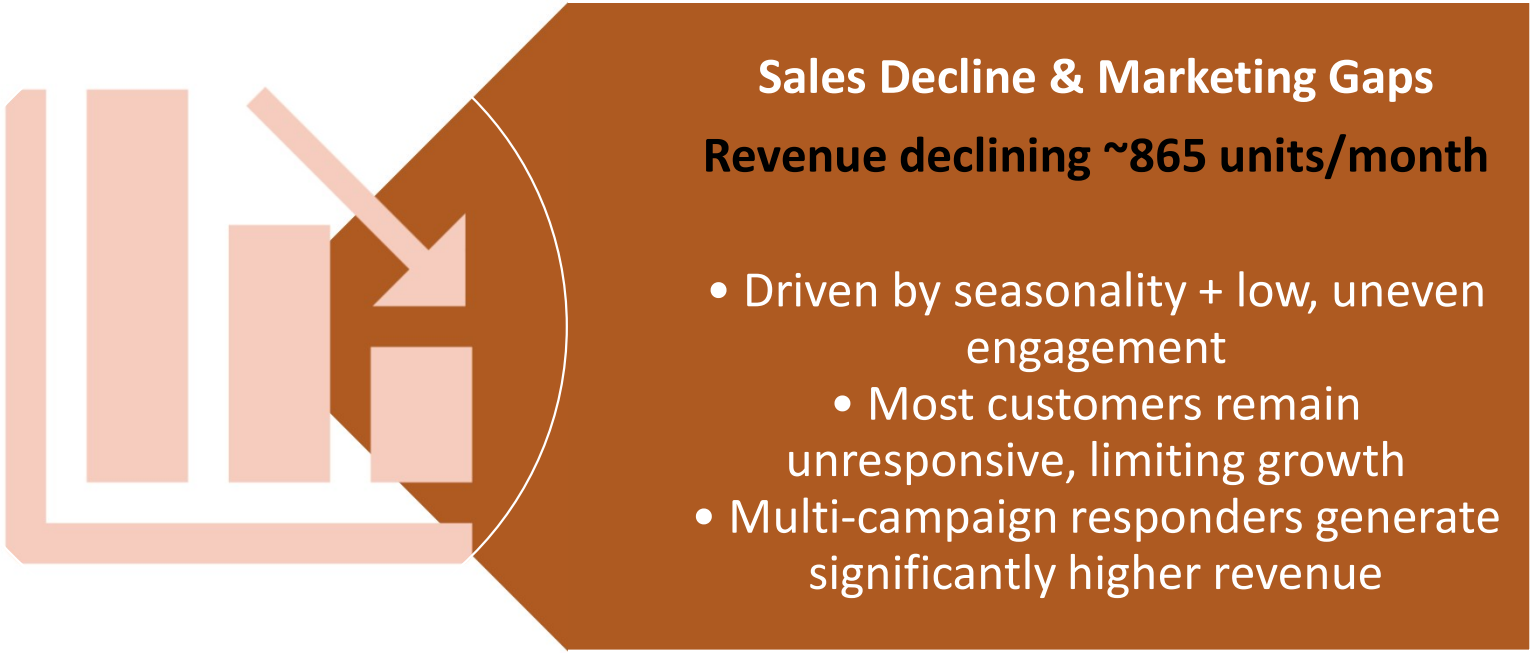
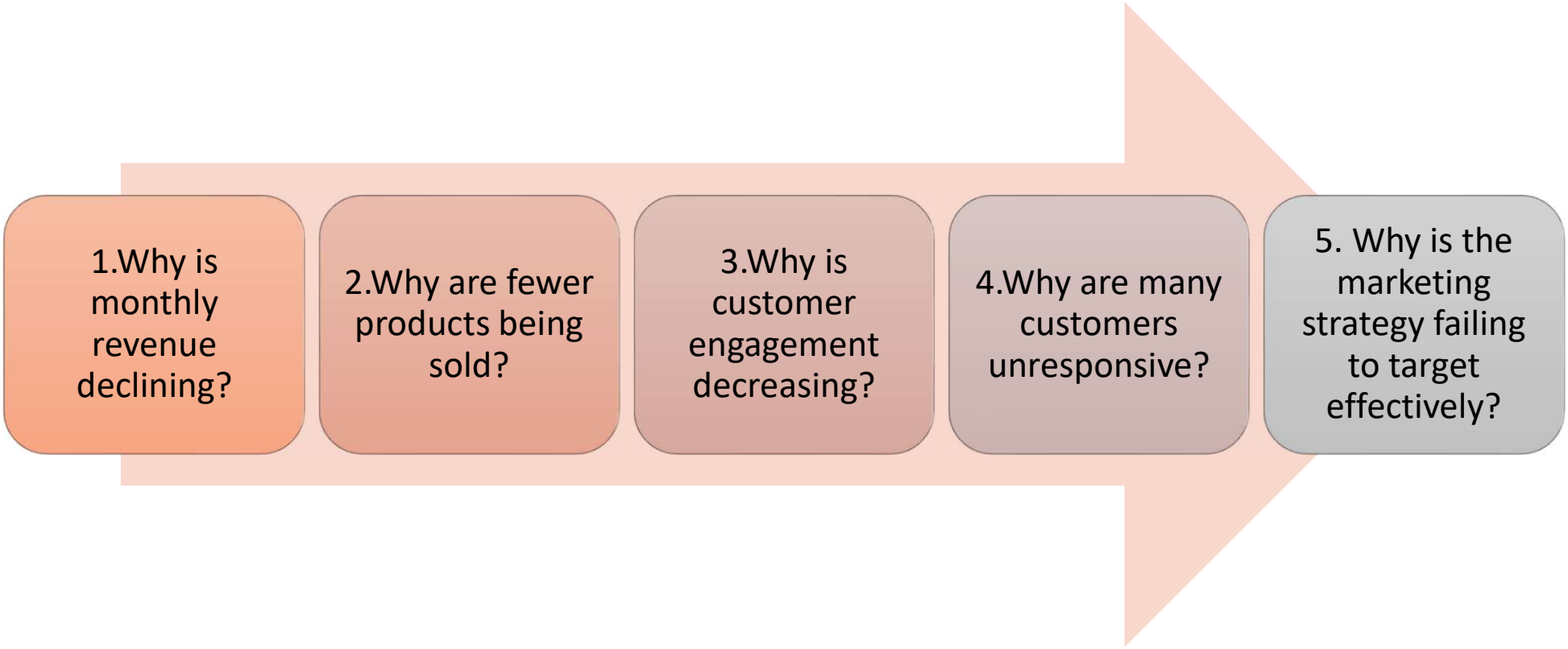
Market Retail Data Analysis

By Lakmali Amarawardana

Background and context of the business



Root Cause Analysis



Customer demography

Age

- Range from 29 - 132 yrs
- Median of 55 mean 56 yrs
- Highest proportion, 50–59: 30.6%
- About 50% of customers are between 48-66 yrs.

Income

Ranges from 1730 - 666666,

Median 51382

Mean 52247

- 50% : earn 35303 -68522
- Lower-Mid: 30,000 – 49,999, 31%

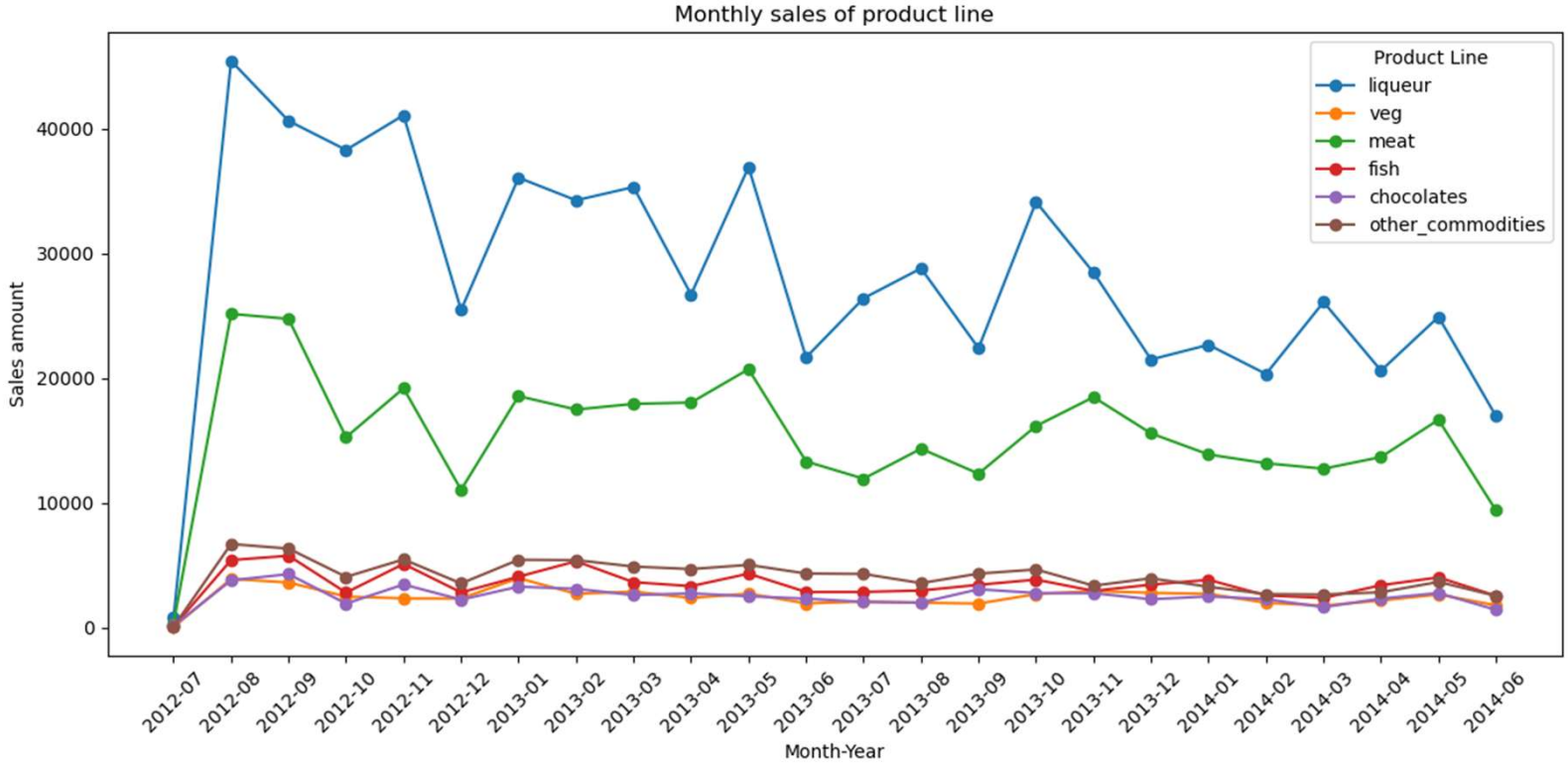
Education levels

- Mostly Graduated, 1116 50.4%

Marital status

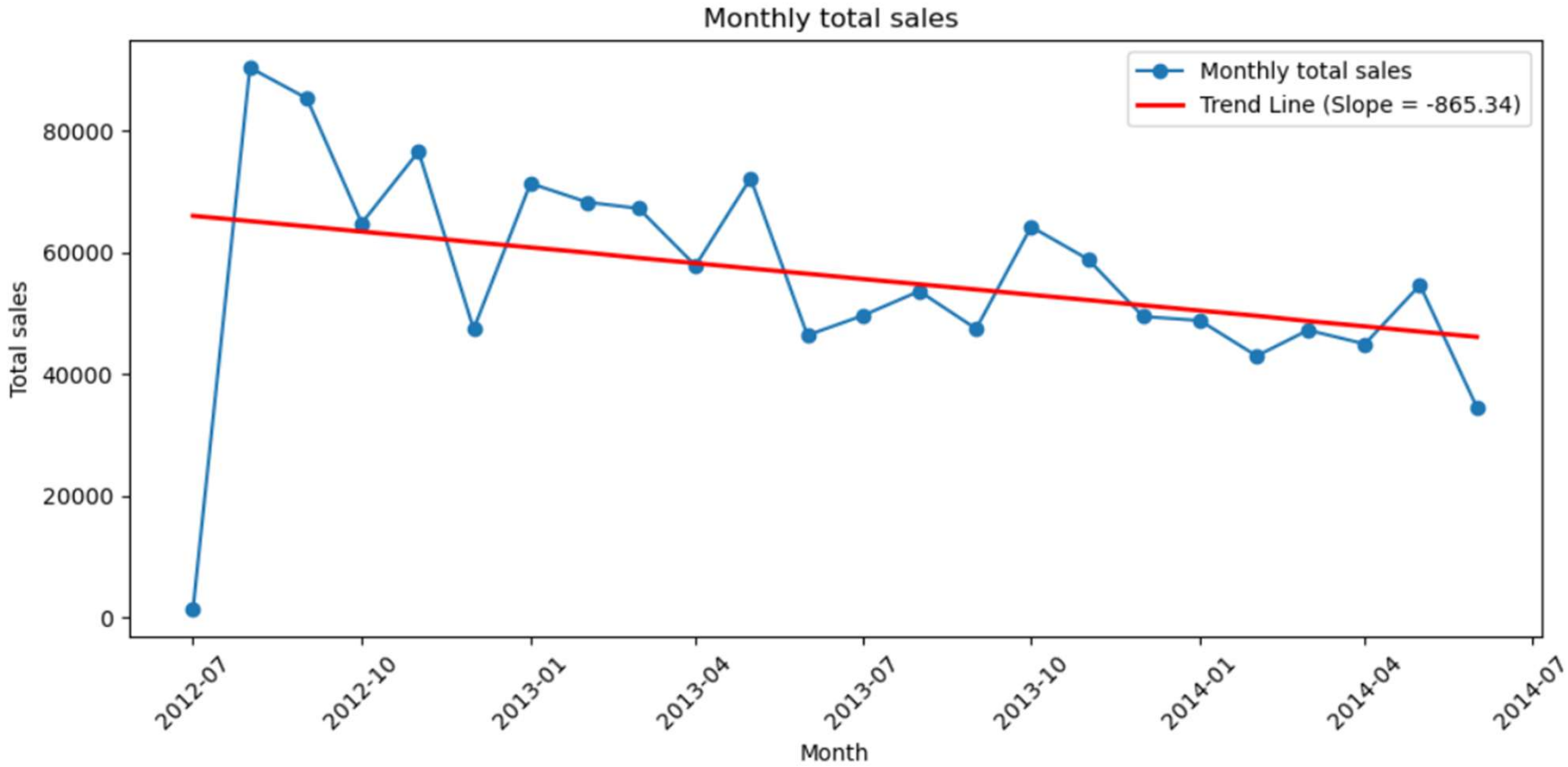
- The majority either Married 38.7% Together 25.86% or Single 21.2%

Monthly sales



**** Monthly sales are on the decline**

Percentage Change from 2012-08 to 2014-06



slope=-865.34

On average, sales are declining by -865 units per month over the analysed period.

Effectiveness of advertising

15% : Ad campaign response rate

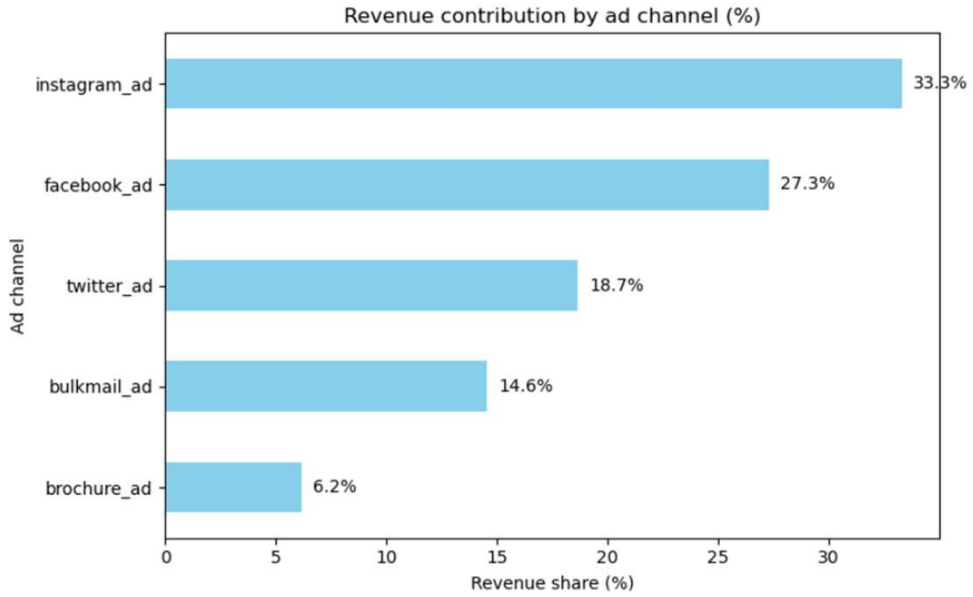
The majority did not respond

Digital channels, Instagram and Facebook generated higher engagement

Responsiveness to ads

- Age: middle-aged customers, 40-69 yrs,11.1%
- Marital status: single, 4.7% and married, 4,2%
- Education level: graduates, 6.9% and PhDs, 4.6%
- Income: middle-to-upper income groups, 13%
- The majority from Spain, 7.9%

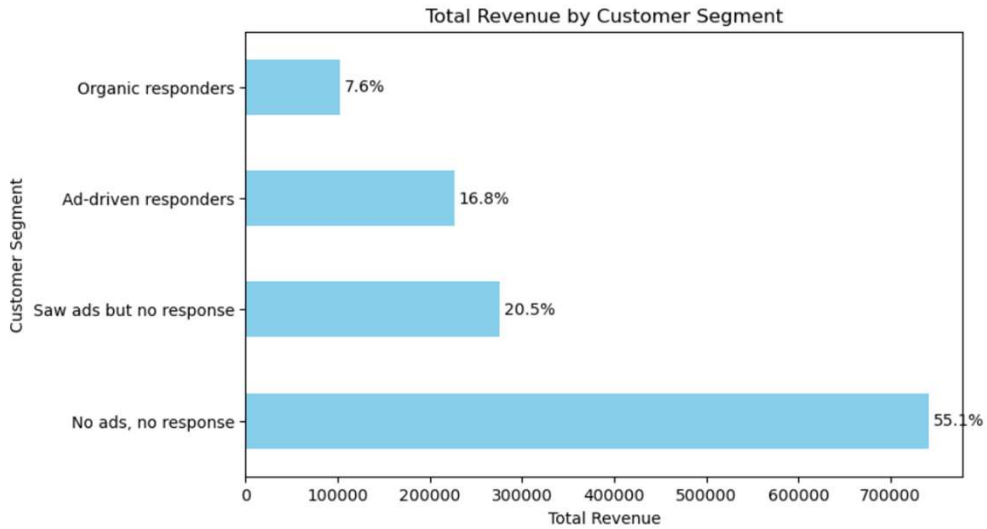
Ad channel effectiveness



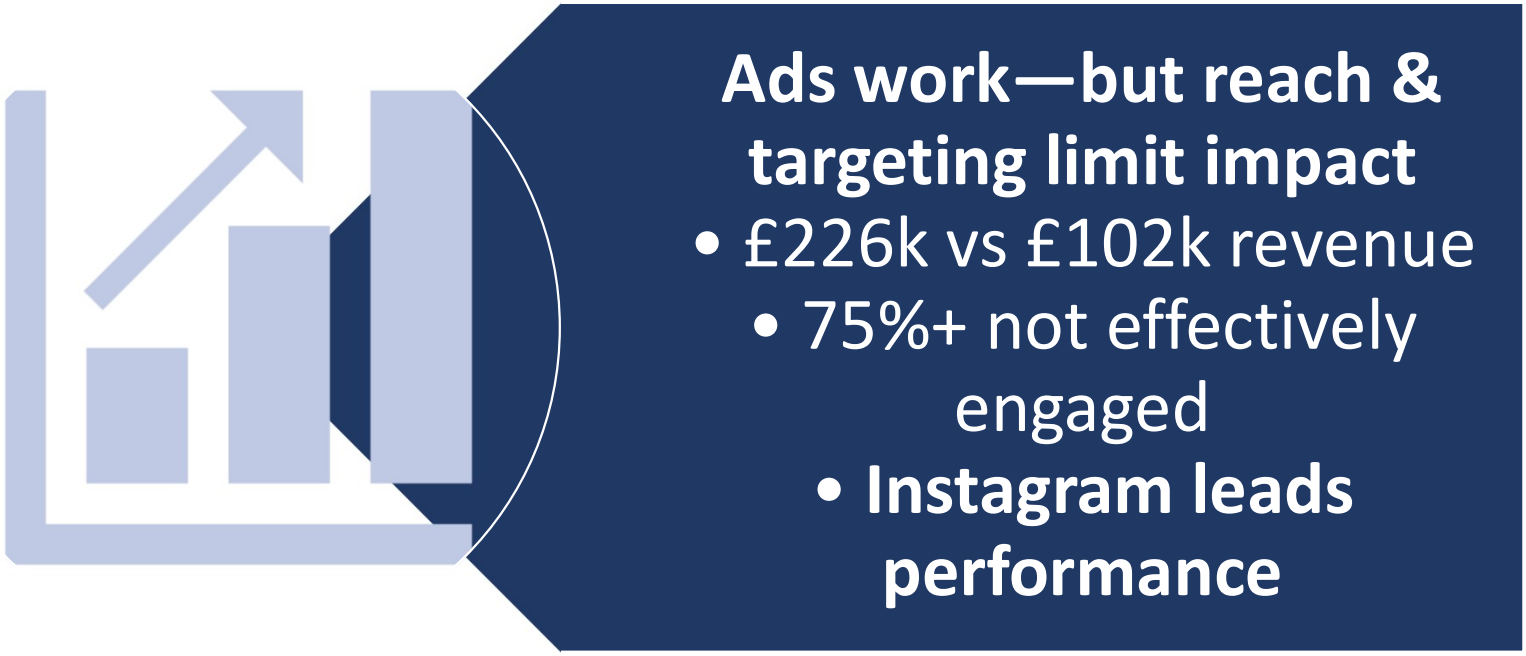
Ad_ch	Revenue	%
instagram_ad	149664	33.3
facebook_ad	122636	27.3
twitter_ad	83805	18.7
bulkmail_ad	65391	14.6
brochure_ad	27657	6.16
Total	449,153	

Segmentation by response to ad campaign

Ads seen	Response	Customer type	Revenue	%
ch_exposure>0	1	Ad_driven responders	226341	16.8
ch_exposure=0	1	Organic responders	101884	7.6
ch_exposure>0	0	saw ads but did not respond	275681	20.5
ch_exposure=0	0	No ads and no response	741373	55.1
		Total	1345279	



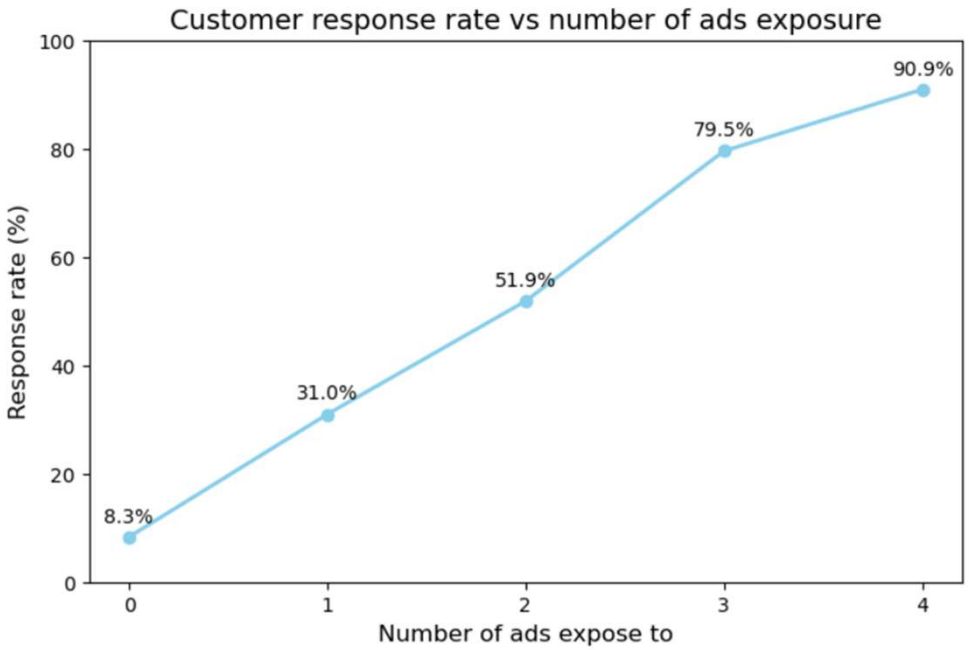
Success of the ad campaign



Revenue generated by each customer segments

- 👉 **No customer in the dataset saw all 5 ads.**

channels_exposed_count	count	sum	response_rate %
0	1757	146	8.3
1	323	100	31.0
2	81	42	51.9
3	44	35	79.5
4	11	10	90.9



Ad Exposure Strongly Increases Response

No ads

- low response (8.3%)

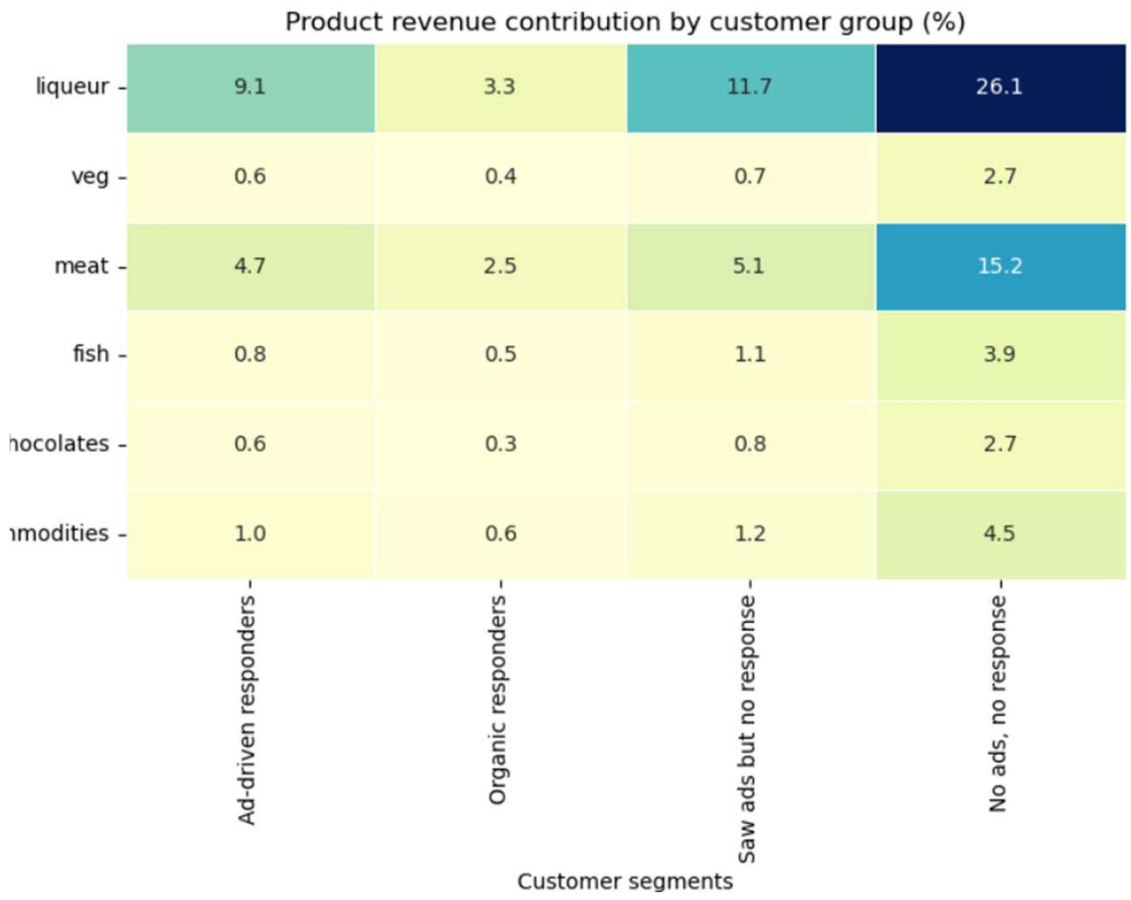
Response rises significantly with more exposure

3 ads: ~79.5% response

4 ads: 90%+ response

Ads Influence Spending, stronger on premium products

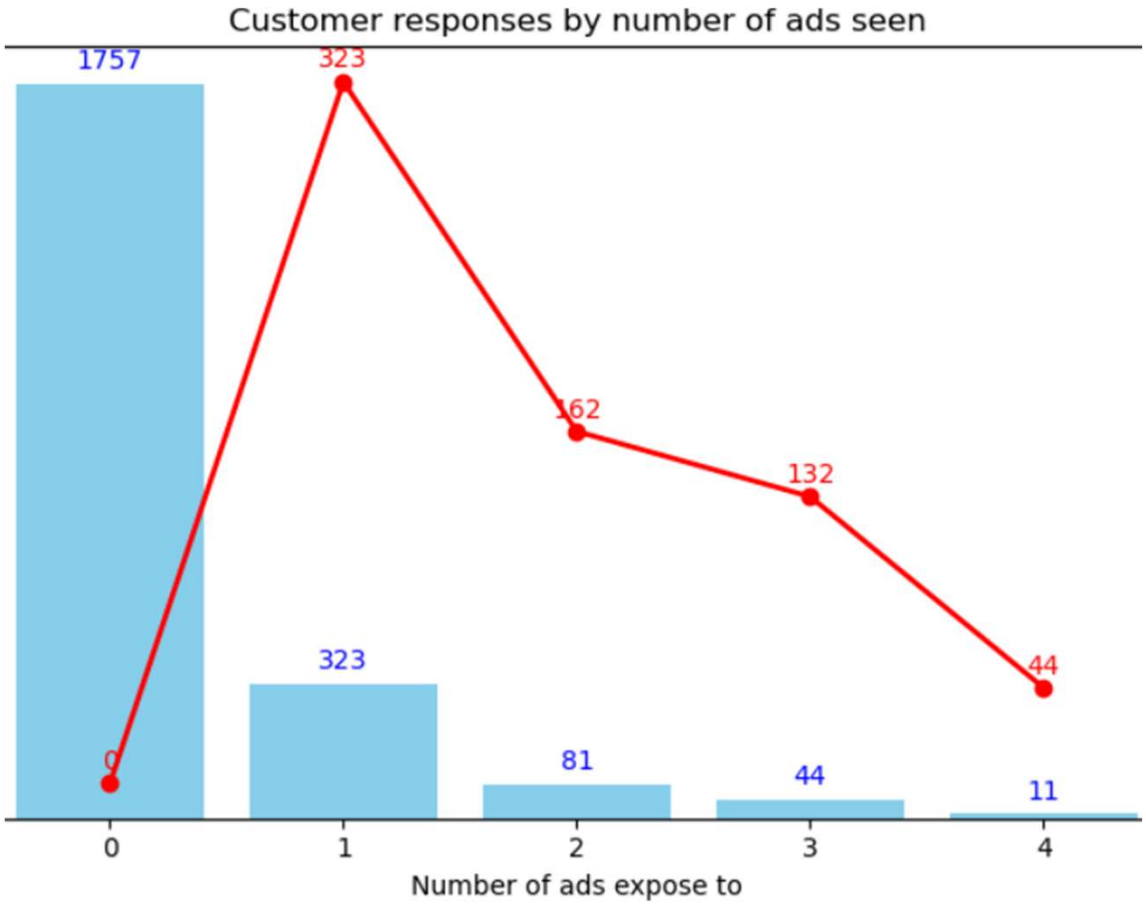
- Highest spending on liqueur & meat across all customer segments.
- Ad increases spending, even without direct response.
- Lower impact on veg, fish & chocolates.
- Ads influence spending unevenly across products.



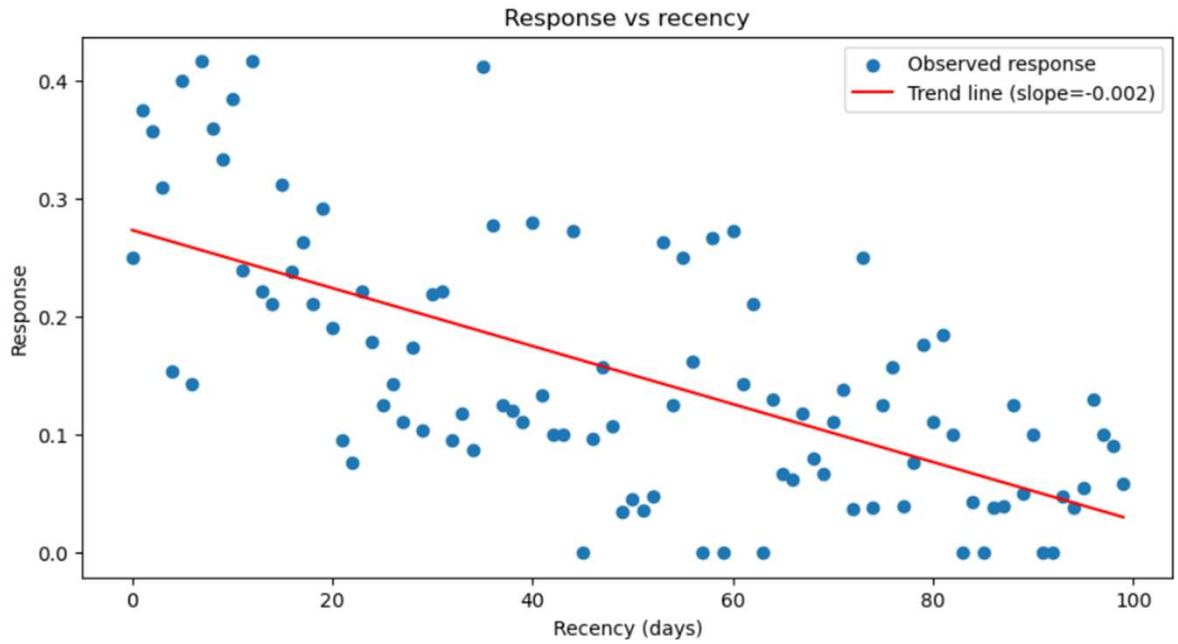
Ad exposure & customer behaviour

- Most customers saw no ads and did not respond.

- Multiple exposures reach fewer customers but generate stronger responses among those reached.



Recency and response



For every 1-day increase in recency, the response decreases by 0.2% (since slope is -0.002 in units of response per day).

Recency and response

Correlation between recency and response: -0.200

The correlation is negative.

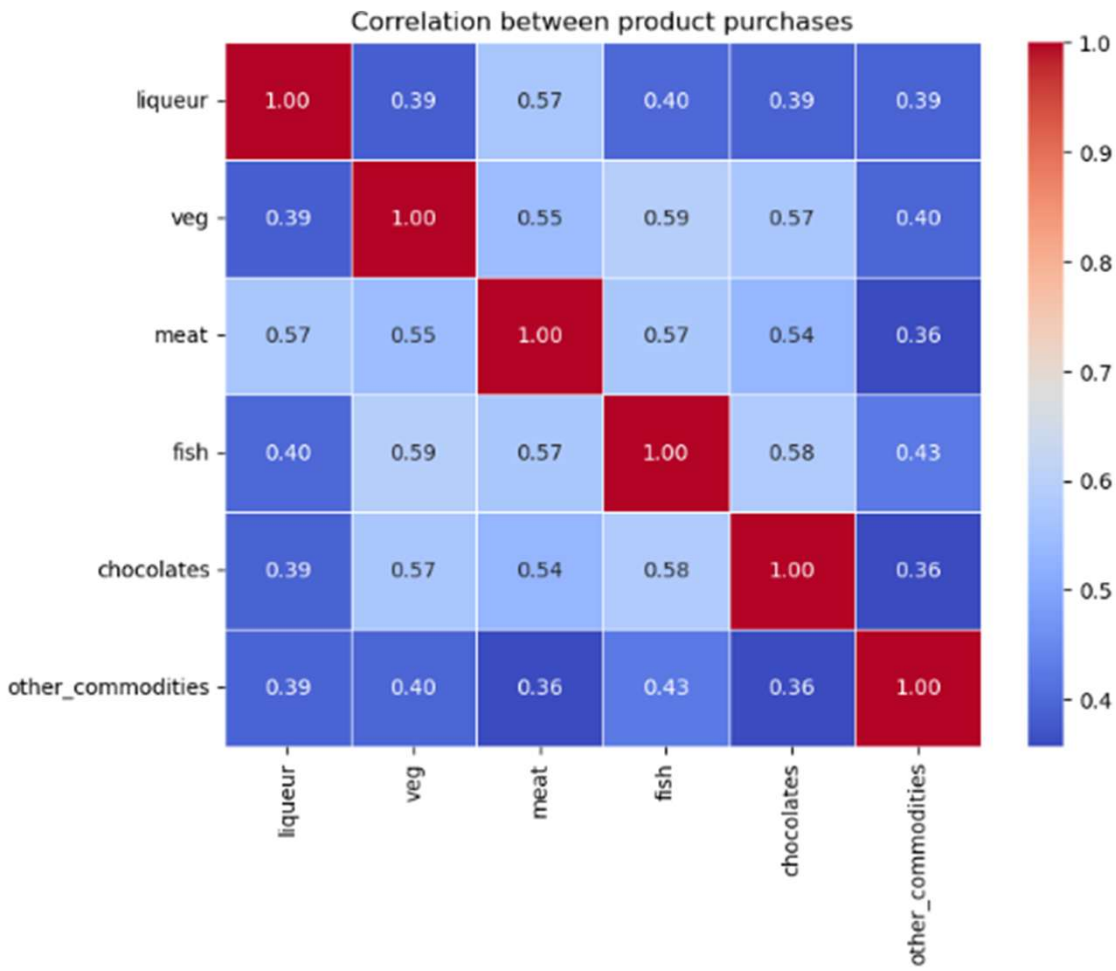
As recency increases (customer hasn’t purchased in a while) the likelihood of response decreases.

Customers who purchased more recently are slightly more likely to respond to a campaign than those who haven’t purchased in a while.

Product purchase relationships (Correlation analysis)

- Strongest link: vegetables & fish 0.59.
- Meat, fish & chocolates: moderate correlation, 0.57–0.58.
- Liqueur & meat moderately correlated, 0.57

- Other products: weaker relationships (0.35–0.43), indicate as independent purchases.

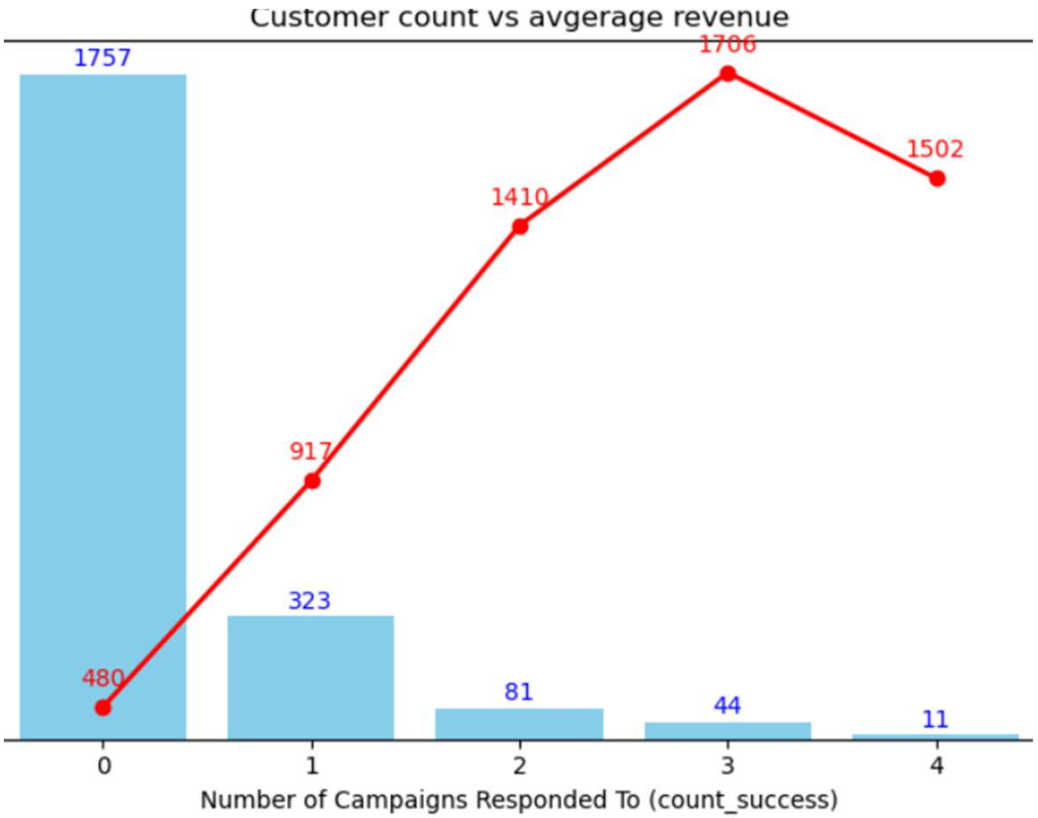


Campaign response & customer value

Strong positive link between campaign engagement and customer value.

More campaign interactions = significantly higher spending.

Engaged customers tend to be higher-value customers.



Campaign Success

country	numwebbuys	response	count_success
Australia	4.05	0.15	0.23
Canada	4.29	0.14	0.33
Germany	4.00	0.15	0.33
India	3.97	0.09	0.26
Montenegro	6.00	0.67*	0.33
South Africa	4.15	0.15	0.26
Spain	4.00	0.16	0.32
United State	4.48	0.12	0.24

income_groups	count_success
Low	0.07
Lower-Mid	0.12
Mid	0.22
Upper-Mid	0.78
High	0.92

As income groups increase from Low to High, the campaign success rises significantly.

*fewer customers

Campaign Success

Web purchases are similar across countries

Response rates vary more significantly

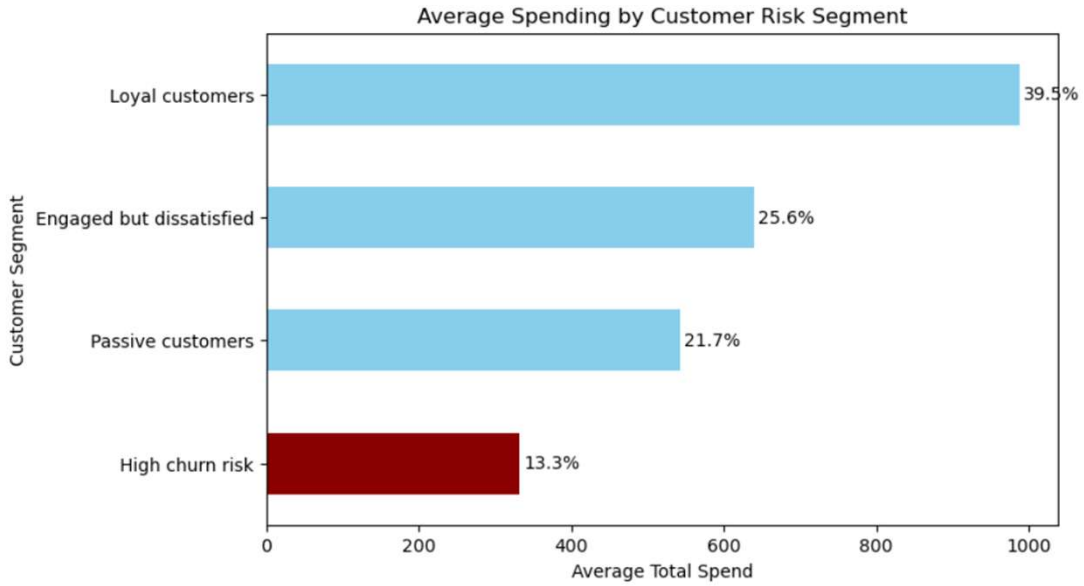
Higher response: Spain, South Africa

Lower response: India, United States

Spain, Canada, Germany show strong campaign success

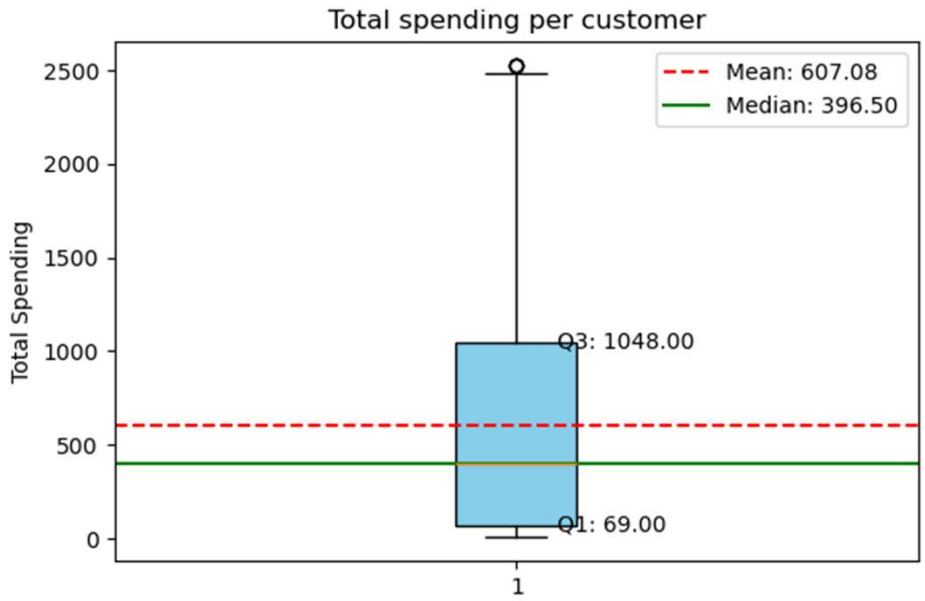
Risk analysis of customer churn

- **Passive customers**: largest group, ~£1.01M revenue, low engagement.
- **Loyal customers**: 330 customers, ~£326k revenue, high value & engagement
- **High churn risk**: 18 customers, ~£6k revenue, low engagement + complaints
- **Engaged but dissatisfied**: 3 customers, ~£1.9k revenue, respond despite issues



High value customers profile

- 553 customers
- Total spending>1048
- Majority are aged 40–69 Predominantly upper-middle income
- Largely concentrated in Spain
- Smaller share from other countries



Conclusion

Low engagement from passive customers is the main issue.

Campaigns work well for a smaller, high-value segment.

Majority of customers remain unresponsive.

Strategic Recommendations

Focus marketing on high-value, responsive segments for greater ROI.

Personalise campaigns using customer data to improve engagement.

Optimise ad frequency to balance reach and intensity.

Explore cross-selling opportunities based on product correlations.

Address concerns of dissatisfied or high-churn customers to prevent revenue loss.

Expand reach, large share of customers are not exposed to ads

Re-engage passive customers with tailored offers or incentives.